Jinrou Li

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EDUCATION

Peking University (PKU) Beijing, China
***B.S. in Life Sciences* (Rank 7/129, GPA 3.79/4.0)** *Sep 2020 – Jul 2024 (expected)*

- **Relevant Curriculum:** Genetics (A, Top3 in class), Genetics Lab(A), Current topics on Genetics(91), Molecular Biology (91), Biochemistry (90), Current topics on Biochemistry(100), Biostatistics (94), Journal Club of the Frontier for Life Sciences (96.5), Selected Readings on Frontiers of Science and Technology (94)

- **Technical Skills:**
- Animal experiments (For mice, embryonic stage and postnatal stage)
  - Perfusion
  - Brain dissection
  - Brain cryosectioning
- Molecular experiments
  - Plasmid construction
  - Genotyping
- Cellular experiments
  - Cell culture (Cell lines: HEK293T, HT1080)
  - Transient transfection (PEI)
  - Confocal imaging
- Biochemistry experiments
  - Western blot
  - Co-immunoprecipitation
  - Luciferase Assay
  - Immunostaining
- Bioinformatic analysis
  - DamID-seq, CUT&TAG, ATAC-seq, RNA-seq
- Programming related
  - Python, R
  - Machine learning (PyTorch)

- **Language:**
- **TOEFL** 106/120 (30/Reading+27/Listening+25/Speaking+24/Writing)
- **GRE** 329/340 (159/Verbal+170/Quantitative+3.5/Analytical Writing)

HONORS & AWARDS

- The Best Oral Presenter of the 9TH symposium for Undergraduate Honor Program in Biology 2022.9
- The third Prize of Peking University Scholarship 2021-2022
- Award for Academic Excellents 2021-2022
- Award for Academic Excellents 2020-2021

RESEARCH EXPERIENCE

Investigating the function of mTF1 in mammalian nervous system Beijing, China
Peking University, School of Life Sciences, Supervisor: Dr. Yan Song *Jan 2022 – present*

- Explored the function of transcription factor mTF1 in the development of mammal nervous system. Discovered the collaboration of mTF1 and histone modifiers in regulating gene expression and found its target genes.

- Accumulated experience in both experiment design and practical skills. Designed a series of experiments for confirming the importance of the collaboration between mTF1 and histone modifiers. Proficient in constructing plasmids for in utero electroporation, Co-immunoprecipitation and confocal imaging. Can do bioinformatic data analysis for DamID-seq and CUT&TAG.

SELECTIVE PROJECT

Purification and enzymatic activity assay of egg white lysozyme Beijing, China
Biochemistry Lab, Peking University *Dec 2021 – Jan 2022*
Chromatography on AKTA to purify proteins, enzymatic activity assay on microplate reader.

SELECTIVE JOURNAL CLUBS

Undergraduate Honor Program in Biology (UHPB) Beijing, China
Of infection and autoimmunity: dual roles $CD8^+$T cells play in human diseases Dec 2021

- $CD8^+$T cells' different phenotypes in cancer and autoimmunity, the crucial biological functions of the balance between the dual roles.

Role of mitochondria in cell fate decisions during neurogenesis May 2022

- Influence of mitochondria morphology and metabolism on neural fate decision, mitochondria associated neural diseases and current treatments.

Mitochondrial DNA base editing tools Nov 2022

- Recent applications of non-CRISPR base editors in mitochondria genome, its future development and therapeutical potential.

Current topics on Biochemistry Beijing, China
Cryo-EM Structure of the Human Ribonuclease P Holoenzyme Oct 2021

- Explanation of the catalyzing mechanism of human ribonuclease P holoenzyme from structure and evolutionary insights.

Current topics on Genetics Beijing, China
Gene technology: Gene-editing tools Mar 2022

- The discovery and development of CRISPR-Cas9 tools, specially focus on base editors and its applications.